Introduction

Feature engineering – creating informative inputs – is often the largest driver of predictive performance. Even deep learning benefits from domain-informed features. Standard techniques: log/sqrt transforms, interactions, polynomial bases, categorical encodings, date features, and lag features for time series.

Prerequisites

Regression / classification basics; understanding the target's scale and distribution.

Theory

Key transformations: - **Log / sqrt** stabilise right-skewed variables. - **Box-Cox / Yeo-Johnson** learn the optimal power transform. - **Polynomial and spline bases** expose non-linear relationships to linear models. - **Interactions** (x_1x_2) surface effects whose direction depends on another variable. - **Target encoding** for high-cardinality categoricals: replace category with smoothed target mean. - **Lag / rolling features** for time series.

Assumptions

Transformations preserve monotonicity where needed; target encoding uses out-of-fold means to avoid leakage.

R Implementation

```r
library(recipes); library(embed)

set.seed(2026)
n <- 300
df <- tibble::tibble(
  x1 = exp(rnorm(n)),                    # right-skewed
  x2 = runif(n, 0, 10),
  cat = sample(letters[1:20], n, replace = TRUE),
  date = as.Date("2024-01-01") + sample(0:364, n, replace = TRUE),
  y = runif(n)
)

rec <- recipe(y ~ ., data = df) %>%
  step_log(x1) %>%
  step_interact(~ x1:x2) %>%
  step_ns(x2, deg_free = 4) %>%
  step_date(date, features = c("dow", "month")) %>%
  step_holiday(date) %>%
  step_rm(date) %>%
  step_lencode_mixed(cat, outcome = vars(y))   # mixed-effects target encoding

baked <- prep(rec) %>% bake(new_data = NULL)
colnames(baked)[1:10]
```

Output & Results

A transformed data frame with log-x1, interaction term, natural spline basis for x2, day-of-week / month features, and smoothly target-encoded category.

Interpretation

"Feature engineering added 12 derived columns: log-transformed skew, spline bases, date features, and target-encoded category. The pipeline is reproducible and applied identically to train and test."

Practical Tips

- Do feature engineering inside a `recipe` (or equivalent) to avoid leakage.
- Log-transform right-skewed variables before linear models; trees handle skew fine.
- High-cardinality categoricals benefit hugely from target encoding (with proper smoothing / CV handling).
- Date features (day-of-week, hour, month, holiday) are often critical for temporal prediction.
- Don't over-engineer – more features means more noise; cross-validate additions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis